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# Effective two-component and two-mediator dark matter

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# Anomaly cancellation of a U(1) gauge symmetry with chiral fermions

In a fundamental theory the elementary fermions are expected to be chiral, i.e., their charges should not allow a mass term larger than the scale of spontaneous symmetry breaking. For any set of charges associated to a U(1) gauge symmetry

$$\mathbf{Z} = [Z_1, Z_2, \dots, Z_N] ,$$

The triangle anomaly with three U(1) gauge bosons on the external lines, i.e., the  $[U(1)]^3$  anomaly, and with one U(1) gauge boson and two gravitons on the external lines should also be cancelled

$$\sum_{\alpha=1}^N Z_{\alpha} = 0, \qquad \sum_{\alpha=1}^N Z_{\alpha}^3 = 0, \qquad (1)$$

## $U(1)_Y$ : no vector-like, i.e., pairs of fields with ‘equal-but-opposite’ charges

The hypercharge for one generation in the standard model can be seen as gauge Abelian symmetry  $U(1)_Y$  with 15 left-handed Weyl massless fermions with integer charges

$$Y = [1, 1, 1, 1, 1, 1, -4, -4, -4, 2, 2, 2, -3, -3, 6]$$

which satisfy

$$\sum_i Y_i = 0, \quad \sum_i Y_i^3 = 0.$$

If we introduce an scalar,  $H$ , of charge 3 we can form Yukawa couplings through the Dirac pairs

$$u_\alpha = (1, -4), (1, -4), (1, -4), \quad d_\alpha = (1, 2), (1, 2), (1, 2), \quad e = (-3, 6)$$

and the *chiral fermions* acquire Dirac masses after the EWSB. They are degenerate because the additional color symmetry. One remains massless,  $\nu_L = (-3)$ . The remnant  $Z_3$  symmetry guarantees the stability of  $e$  and the other lightest quark,  $d_\alpha$  is accidentally stable.

The two mediators of the interactions are  $B_\mu$  and  $H$ .

## Secluded gauge $U(1)_D$ without vector-like fermions:

$$\mathbf{S} = [\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_{N'}]$$

- *Dark Higgs mechanism*: Singlet dark Higgs  $\phi$  acquires a vev and give mass to the dark photon

$$\mathcal{L} = i\psi_a^\dagger \overline{\sigma}^\mu \left( \partial_\mu - ig_D Z_\mu^D \right) \psi_a - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{a < b} h_{ab} \psi_a \psi_b \phi^{(*)} + \text{h.c.} - V(\phi). \quad (2)$$

- $z_\alpha$  are the charges of SM-singlet right-handed chiral fermions with  $N \geq 5$ 
  - $\chi_i$  *massless fermions* with  $i = 1, \dots, N'$  with  $N' \leq N$
  - $\psi_a$  *multi-component dark matter*: massive after the spontaneous symmetry breaking of  $U(1)_D$  with  $a = N' + 1, \dots, N$
- *Larger parameter space*: Dark photon,  $Z_D$ , exclusions instead of  $Z'$

Decrease the number of charges to be assigned to dark matter particles,  $\psi_i$  below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \dots, \psi_N]$$

$$\chi_1 \rightarrow \nu_{R1}, \dots, \chi_{N_\nu} \rightarrow \nu_{RN_\nu}, \quad 2 \leq N_\nu \leq 3,$$

$$\mathcal{L}_{\text{eff}} = h_{\nu}^{ij} (\nu_{Ri})^\dagger \epsilon_{ab} L_j^a H^b \left( \frac{\phi^*}{\Lambda} \right)^\delta + \text{H.c.}, \quad \text{with } i, j = 1, 2, 3,$$

$\phi$  is dark Higgs responsible for the SSB of the anomaly-free gauge symmetry and

give mass to all  $\psi_a$ .  $\nu_R$  remains massless

$$\phi = -\frac{\nu}{\delta},$$

Multi-component dark matter model with two-mediators:  $Z_D$  and  $\phi$

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Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \rightarrow \nu_{R1}, \chi_2 \rightarrow \nu_{R2}, \quad N_\nu = 2,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} (\nu_{Ra})^\dagger \epsilon_{bc} L_j^b H^c \left( \frac{\phi^*}{\Lambda} \right) + \text{H.c.}, \quad \text{with } j = 1, 2, 3,$$

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Two-component dark matter model with two-mediators:  $Z_D$  and  $\phi$



# Minimal secluded model with D-5 effective Dirac neutrino masses

We assume decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\psi_i^\dagger \not{D} \psi_i - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{i < j} h_{ij} \psi_i \psi_j \phi^{(*)} + \text{h.c.} - V(\phi).$$

two-component and two-mediator DM with two Dirac-fermion DM particles

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two-component and two-mediator DM with two Dirac-fermion DM particles

$$z = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)]$$

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two-component and two-mediator DM with two Dirac-fermion DM particles

$$z = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)]$$

$$\mathcal{L} \subset h_{(-3, -2)} \psi_{-3} \psi_{-2} \phi^* + h_{(1, -6)} \psi_1 \psi_{-6} \phi^* + \text{h.c.}$$

# Minimal secluded model with D-5 effective Dirac neutrino masses

We assume *a minimal* decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\psi_i^\dagger \not{D} \psi_i - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{i < j} h_{ij} \psi_i \psi_j \phi^{(*)} + \text{h.c.} - V(\phi).$$

two-component, two-mediator DM with two-flavor scotogenic sector

$$\mathbf{z} = [-9, -9, -9, 10, 10, -1, -1, 4, 5] \rightarrow \phi = 9 \rightarrow [(-9, -9, -9), (10, -1), (10, -1), (4, 5)]$$

$$\mathcal{L} \subset \underbrace{\sum_{a,b=1}^2 h_{(10a,-1b)} \psi_{10} \psi_{-1} \phi}_{\text{multi-flavor scotogenic DM}} + h_{(4,5)} \psi_4 \psi_5 \phi^* + \text{h.c.} \quad (3)$$

multi-flavor scotogenic DM

September 24, 2021

Dataset

Open Access

# Set of N integers between -30 and 30 with sum and cubic sum up to zero for $4 < N < 13$

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## Anomalies

Solutions obtained with the python package: [anomalies](#) based on the method to find anomaly free solutions of the standard model extended with an Abelian Dark Symmetry with  $N$  right-handed singlet chiral fields described in [arXiv:1905.13729](#) [PRL].

## Data scheme

- 'l': integer lists → input to obtain the 'solution' by using the [anomalies](#) package
- 'k': integer lists → input to obtain the 'solution' by using the [anomalies](#) package
- 'solution': list → of integers,  $z_i$  which satisfy  $\sum_{i=1}^N z_i = 0$  and  $\sum_{i=1}^N z_i^3 = 0$ .
- 'n': integer → number of integers in 'solution',  $N$ .

## USAGE

```
#Example of JSON file usage in Python with pandas (see also json module)
>>> import pandas as pd
>>> df=pd.read_json('solutions.json.gz')
>>> df[:2]
   l      k      solution gcd n
0  [1, 2] [0, -3] [1, 5, -7, -8, 9] 1 5
1  [-2, -1] [0, -1] [2, 4, -7, -9, 10] 1 5
```

## Data:

2 296 615 solutions with  $5 \leq N \leq 12$  integers until 'j32' [JSON]

141

views

351

downloads

[See more details...](#)

Indexed in

OpenAIRE

## Publication date:

September 24, 2021

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DOI [10.5281/zenodo.7380817](#)

## Keyword(s):

Anomaly free Diophantine equations Abelian symmetry  
Gauge Symmetry

## License (for files):

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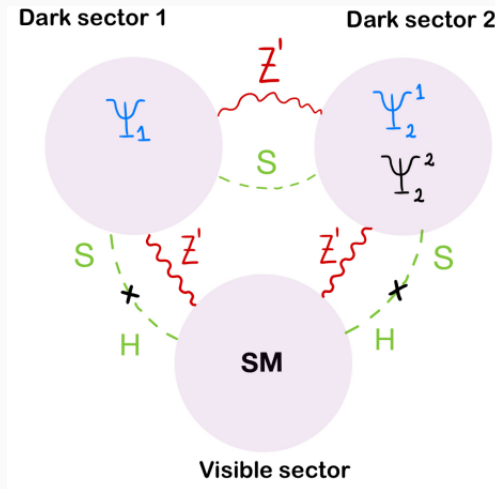
## Versions

Version v2

Sep 24, 2021

[10.5281/zenodo.7380817](#)

| Solutions $\sim 1000$                     | $N$ | $\nu$ | $\delta$ | $\phi$ | $N_D$ | $N_M$ | $G_D$ | $G_M$ |
|-------------------------------------------|-----|-------|----------|--------|-------|-------|-------|-------|
| (5, 5, -2, -3, 1, -6)                     | 6   | 5     | 1        | -5     | 2     | 0     | 1     | 0     |
| (-9, -9, -9, 10, 10, -1, -1, 4, 5)        | 9   | 9     | 1        | -9     | 3     | 0     | 2     | 0     |
| (1, -2, 3, 4, 6, -7, -7, -7, 9)           | 9   | -7    | 1        | 7      | 3     | 0     | 1     | 0     |
| (1, -2, -2, 3, 3, -4, -4, 6, 6, -7)       | 10  | 6     | 1        | -6     | 3     | 2     | 2     | 2     |
| (1, -2, -2, 3, 4, -5, -5, 7, 7, -8)       | 10  | -5    | 1        | 5      | 4     | 0     | 2     | 0     |
| (1, -2, -2, 3, 5, -6, -6, 8, 8, -9)       | 10  | -6    | 1        | 6      | 4     | 0     | 2     | 0     |
| (2, 2, 3, 4, 4, -5, -6, -6, -7, 9)        | 10  | 2     | 1        | -2     | 4     | 2     | 2     | 2     |
| (1, 1, 5, 5, 5, -6, -6, -6, -9, 10)       | 10  | 1     | 1        | -1     | 4     | 0     | 3     | 0     |
| (2, 2, 4, 4, -7, -7, -9, -9, 10, 10)      | 10  | 10    | 2        | -5     | 3     | 0     | 2     | 0     |
| (1, 2, 2, -3, 6, 6, -8, -8, -9, 11)       | 10  | -8    | 1        | 8      | 4     | 1     | 2     | 1     |
| (1, -2, -3, 5, 6, -8, -9, 11, 11, -12)    | 10  | 11    | 1        | -11    | 4     | 0     | 1     | 0     |
| (1, 1, -3, 4, 4, -7, 8, -10, -10, 12)     | 10  | -10   | 2        | 5      | 4     | 0     | 2     | 0     |
| (1, 1, -2, -2, -4, 6, -10, 11, 12, -13)   | 10  | -2    | 1        | 2      | 3     | 2     | 1     | 2     |
| (3, 4, 4, 4, 4, -5, -8, -8, -11, 13)      | 10  | -8    | 1        | 8      | 2     | 4     | 1     | 4     |
| (4, 4, 5, 6, 6, -9, -10, -10, -11, 15)    | 10  | 6     | 1        | -6     | 4     | 0     | 2     | 0     |
| (1, -2, -4, 7, 7, -10, -12, 14, 14, -15)  | 10  | 14    | 1        | -14    | 3     | 2     | 1     | 2     |
| (1, 2, 2, -3, 4, -6, 12, -13, -14, 15)    | 10  | 2     | 1        | -2     | 4     | 1     | 1     | 1     |
| (1, 4, 4, -7, 8, 8, -9, -12, -12, 15)     | 10  | 8     | 1        | -8     | 4     | 2     | 2     | 2     |
| (1, 2, 2, -9, -9, 16, 16, 17, -18, -18)   | 10  | -18   | 1        | 18     | 3     | 2     | 2     | 2     |
| (1, -3, -6, 7, -10, 11, -16, 18, 18, -20) | 10  | 18    | 2        | -9     | 4     | 0     | 1     | 0     |
| (1, -4, 5, -6, -6, 10, -14, 15, 20, -21)  | 10  | -6    | 1        | 6      | 4     | 0     | 1     | 0     |
| (2, -3, -6, 7, 12, -14, -14, 17, 20, -21) | 10  | -14   | 1        | 14     | 4     | 1     | 1     | 1     |
| (3, 6, 6, -7, 8, 8, -14, -14, -17, 21)    | 10  | -14   | 1        | 14     | 4     | 1     | 2     | 1     |
| (8, 8, 9, 10, 10, -13, -18, -18, -27, 31) | 10  | -18   | 1        | 18     | 4     | 1     | 2     | 1     |



Decoupled inert scalar sector  $\rightarrow \Delta N_{\text{eff}} = 0$

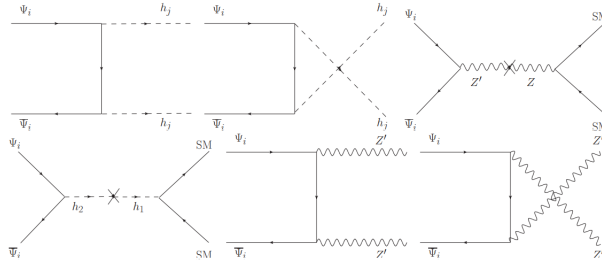
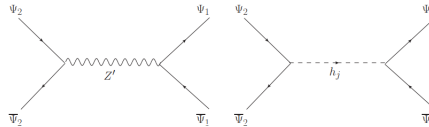


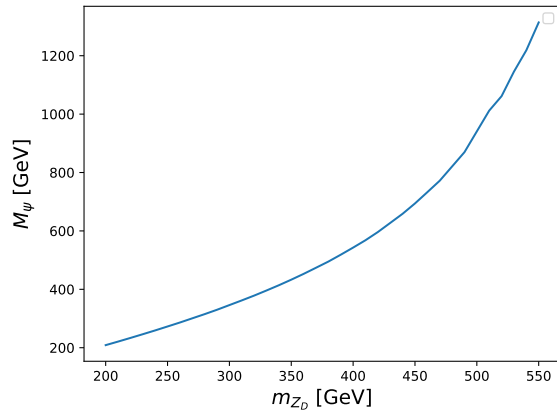
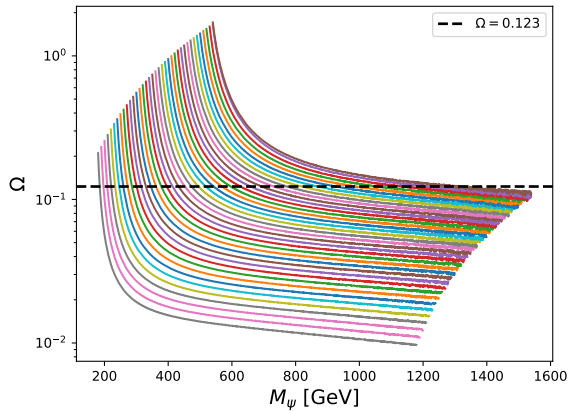
Figure 3: DM annihilation channels.


Figure 4: DM conversion channels  $\Psi_1 \leftrightarrow \Psi_2^i$ .

Decoupled inert scalar sector  $\rightarrow \Delta N_{\text{eff}} = 0$



$M_2 \gg M_1$  and  $g_D = 0.1$



# Conclusions

